

Hybrid ML-SLM RAG System for Large Technical PDFs

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Abstract—We present a novel hybrid Retrieval-Augmented Generation (RAG) system designed specifically for processing and extracting information from large technical PDFs. Our system addresses the challenge of efficiently managing extensive technical documentation by combining traditional machine learning approaches with Small language models. The architecture employs a dual-retrieval mechanism that leverages both TF-IDF vectorization and dense neural embeddings to optimize information retrieval accuracy from complex technical content. The system features an advanced PDF processor with technical content-aware chunking strategies that preserve the integrity of specialized content such as mathematical formulas and section hierarchies. Memory efficiency is achieved through intelligent batch processing and optimized vector storage using FAISS when available. For generation, we integrate Google's Gemma 3 1b got Highest accuracy 68+ and other SLM with failover mechanisms to ensure consistent output quality. Empirical evaluation demonstrates that our hybrid approach outperforms single-method retrieval systems when handling specialized technical terminology and complex conceptual relationships. The system successfully processes collections of 8-15 large PDFs (500+ pages each) while maintaining contextual relevance in responses. This approach represents a significant advancement in making large technical document collections more accessible and queryable through natural language, with particular relevance for scientific research, technical documentation management, and educational Books applications.

Index Terms—TF-IDF, RAG, SLM,

I. INTRODUCTION

The general availability of technical and scientific literature in digital formats has brought about previously unheard-of access to knowledge, but it has also brought about substantial computational difficulties. Using standard natural language processing techniques to query and extract information from technical documentation is still challenging, especially when it comes to large PDF files that contain mathematical notations, domain-specific vocabulary, and intricate hierarchical structures. Managing collections of lengthy texts (500+ pages apiece) that include related ideas from several sources makes this problem very severe.

Conventional information retrieval systems have mostly relied on statistical techniques like TF-IDF¹ and keyword matching, which frequently fall short of capturing the contextual relationships and semantic subtleties found in technical literature. The use of dense neural embeddings in more recent methods has enhanced semantic understanding², but it may overlook clear terminological matches that are crucial in technical areas. Large Language Models (LLMs) have shown impressive ability in natural language generation and processing, however when utilized alone, they have limits related to knowledge recency and hallucinations.

Retrieval-Augmented Generation (RAG)³ systems have emerged as a promising approach to mitigate these limitations by grounding LLM responses in retrieved context. However, existing RAG implementations face several challenges when applied to large technical documents: Most RAG systems struggle with the scale and complexity of large technical PDFs, leading to prohibitive processing times or memory requirements. Generic text chunking strategies often disrupt the coherence of technical content such as mathematical equations, algorithms, and structural hierarchies.

Small Language Models (SLMs)⁴ are compact, efficient versions of traditional language models, typically ranging from 1 million to 10 billion parameters, designed to operate in resource-constrained environments while retaining core NLP capabilities like text generation, summarization, and translation. These models are created through techniques such as knowledge distillation, pruning, and quantization, balancing performance with reduced computational demands. Examples include Meta's Llama3.2-1B, Microsoft's Phi-3.5-Mini-3.8B, and Google's Gemma3-4B. SLMs offer advantages like lower energy consumption, faster inference, and on-device deployment but face limitations in scope and robustness compared

¹<https://www.geeksforgeeks.org/understanding-tf-idf-term-frequency-inverse-document-frequency/>

²<https://www.cloudflare.com/learning/ai/what-are-embeddings/>

³<https://aws.amazon.com/what-is/retrieval-augmented-generation/>

⁴<https://huggingface.co/blog/jjokah/small-language-model>

to larger models. Their applications span chatbots, edge computing, healthcare, and education, making AI more accessible and customizable.

We show experimentally that this strategy performs much better than single-method retrieval systems for specialized technical documentation. By processing collections of 8–15 big PDFs with success and preserving contextual relevance in responses, the system makes it possible to extract knowledge from technical literature more effectively.

To address these challenges, we present a novel hybrid ML-LLM RAG system specifically designed for processing large technical PDFs. Our system makes several key contributions:

- 1) We introduce an advanced PDF processor with technical content-aware chunking strategies that preserve the integrity of specialized content while maintaining computational efficiency.
- 2) We develop a hybrid retrieval approach that synergistically combines TF-IDF vectorization with dense neural embeddings to optimize retrieval performance for technical content.
- 3) We implement memory-efficient batch processing techniques and adaptive fallback mechanisms that enable the system to function across a range of computational environments.
- 4) We integrate the Gemma 3 and Many other Small language model with specialized prompting and error-handling protocols to ensure high-quality responses grounded in the retrieved document context.

The remainder of this paper is organized as follows: Section II Literature reviews in document processing, information retrieval, and retrieval-augmented generation. Section III details the architecture and components of our hybrid system. Section IV presents our experimental Result and evaluation results. Section V discusses Conclusion and future directions.

II. LITERATURE REVIEW

Reciprocal rank fusion and dynamic chunking are two improvements that were introduced in this paper [1] to increase retrieval accuracy for context augmentation. In order to achieve better benchmarks on datasets such as SQUAD and TREC-COVID, Sawarkar et al. [2] significantly improved RAG by creating a "Blended RAG" technique that combined hybrid query tactics with semantic search. Researchers [3] presented a hybrid RAG approach that combined the Sentence-Window and Parent-Child approaches, showing notable gains in fidelity and contextual appropriateness. In a similar vein, Chuang and Chen [4] highlighted how RAG can be deployed through a hybrid cloud architecture, improving flexibility, scalability, and privacy. HyPA-RAG was first presented by Kalra et al. [5] and is designed for AI legal and policy environments. It uses hybrid retrieval techniques and adaptive parameter adjustment to greatly increase response fidelity. Applying sophisticated diffusion-based reinforcement learning algorithms, this research [7] integrated hybrid RAG into multi-modal LLMs for safe and effective healthcare data management inside the Internet of Medical Things (IoMT). In hybrid RAG

systems, the authors [8] created extensive improvements for complicated reasoning tasks, including lowering hallucinations and enhancing numerical reasoning skills. RAG-FW, a hybrid convolutional framework designed for automated extraction and grading of retinal lesions, was previously presented by Hassan et al. [6], highlighting RAG's broad range of applications outside NLP. [10] Using data from social media, including PDFs, this work investigates how to optimize Large Language Models (LLMs) for sentiment analysis on AI-related subjects.

Ask-EDA, an LLM-powered assistant that combines hybrid RAG and abbreviation de-hallucination (ADH), was created by Shi et al. [17] and greatly increases recall and accuracy in electronic design jobs. A hybrid RAG-NLP system named CareerBoost was used in recruitment by Deshmukh and Bajaj [9], which improved the relevancy of job recommendations and raised user engagement and application rates. Through simulation experiments, Zhu, Nguyen, and Cripps [20] demonstrated the superiority of their "Run it As Gaussian" (RAG) method over conventional discretization techniques for handling hybrid Bayesian networks. Su et al. [18] used diffusion-based contract theory to maximize data security and freshness in their safe hybrid RAG-powered multimodal LLM framework for healthcare data management. A hybrid RAG and In-Context Learning (ICL) approach was investigated by researchers [12], which greatly improved the contextual relevance of automated question production. Shah-Mohammadi and Finkelstein [16] achieved great precision and recall by extracting drug use information from clinical notes using a RAG-GPT-4 pipeline. [13] To create adversarial examples, researchers combined three independent attacks into two ensemble adversarial attacks. Hybrid robot-assisted gait training (RAGT) was studied by the authors [11], however there was little proof that it was better than traditional rehabilitation for stroke recovery. RAG Playground, a systematic evaluation framework that highlights the significance of retrieval techniques and prompt engineering in RAG systems, was introduced by Papadimitriou et al. [15]. A thorough analysis of assessment measures and benchmarks for evaluating RAG model performance was presented in this work [19], which also highlighted important issues and future directions. Mandikal [14] investigated retrieval-augmented LLMs designed for specialized fields like ancient Indian philosophy, and they outperformed conventional LLMs in terms of factual accuracy and less hallucinations.

III. METHODOLOGY

Our hybrid ML-SLM RAG system for processing large technical PDFs. The system consists of four main components: (1) PDF processor, (2) hybrid retriever, (3) SLM generator, and (4) integration framework. Figure 1 illustrates the overall architecture. We share our code in Github Repo ⁵ Also for other authors.

A. Advanced PDF Processing

Our PDF processing pipeline transforms large technical documents into semantically coherent chunks while preserving

⁵https://github.com/rafi79/HybridRAGML_LLM

Hybrid ML-LLM RAG System for PDF Processing

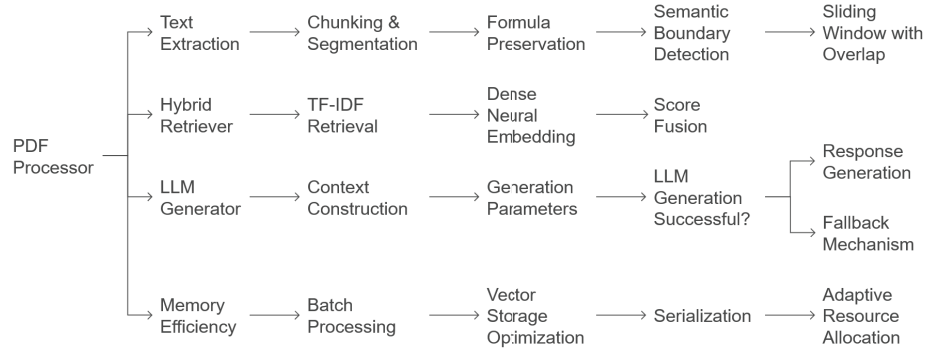


Fig. 1. Hybrid ML-LLM RAG System

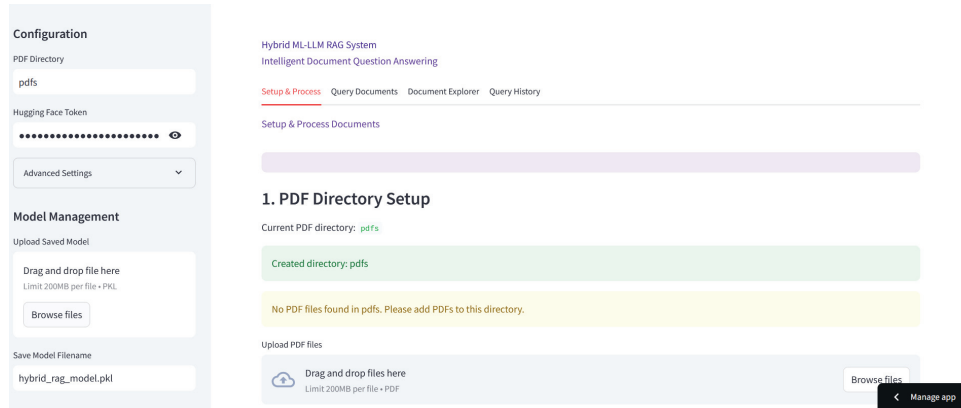


Fig. 2. Hybrid ML-SLM RAG System Streamlit Interface

the integrity of specialized content such as mathematical formulas and section structures.

1) *Text Extraction*: We employ PyPDF2 for text extraction with optimizations for large documents:

- Page-by-page processing to minimize memory usage
- Strategic garbage collection at regular intervals
- Filtering of mostly empty pages

2) *Technical Content-Aware Chunking*: The core innovation in our chunking approach is the preservation of technical content integrity:

- 1) *Section-Based Segmentation*: We identify section boundaries in technical documents using pattern matching to detect headings, numbered sections, and other structural markers.
- 2) *Formula Preservation*: Mathematical formulas are temporarily replaced with placeholder tokens during processing:

$$\text{text}_{\text{processed}} = \text{Replace}(\text{text}_{\text{original}}, \text{formulas}, \text{placeholders}) \quad (1)$$

B. Hybrid Retrieval System

Our hybrid retrieval system combines the strengths of statistical and neural approaches to optimize retrieval from technical documents.

1) *TF-IDF Retrieval*: The TF-IDF component employs several enhancements for technical content:

- N-gram representation (unigrams and bigrams)
- Domain-specific preprocessing that preserves technical terminology
- Optional dimensionality reduction using Truncated SVD

For a query q and document chunk c , the TF-IDF similarity is calculated as:

$$\text{sim}_{\text{TF-IDF}}(q, c) = \frac{\vec{q}_{\text{TF-IDF}} \cdot \vec{c}_{\text{TF-IDF}}}{\|\vec{q}_{\text{TF-IDF}}\| \|\vec{c}_{\text{TF-IDF}}\|} \quad (2)$$

2) *Dense Neural Embedding*: The dense retrieval component utilizes the all-MiniLM-L6-v2⁶ model to generate semantic vector representations:

- Batch processing for memory efficiency

⁶<https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2>

- Vector normalization for cosine similarity calculations
- Fallback to TF-IDF with SVD when sentence transformers are unavailable

3) *Score Fusion*: The hybrid retriever combines results from both methods using a weighted approach:

$$\text{sim}_{\text{hybrid}}(q, c) = \alpha \cdot \text{sim}_{\text{TF-IDF}}(q, c) + (1 - \alpha) \cdot \text{sim}_{\text{dense}}(q, c) \quad (3)$$

where α is the weighting parameter (default 0.3).

This fusion leverages the complementary strengths of both approaches: TF-IDF's precision with technical terminology and dense embedding's semantic understanding capabilities.

C. LLM Generator

The generation component utilizes the Gemma 3 language model with several enhancements for robustness in technical contexts.

1) *Context Construction*: Retrieved document chunks are formatted into a structured prompt:

- Each chunk includes source attribution (document name and page number)
- The system is instructed to use only information from provided documents
- Explicit instructions to acknowledge information gaps when present

The prompt structure can be represented as:

$$\text{Prompt} = \text{Instruction} + \sum_{i=1}^k \text{Format}(c_i, \text{source}_i, \text{page}_i) + \text{Query} \quad (4)$$

2) *Fallback Mechanism*: To ensure system reliability, we implement a fallback extractive summarization approach that activates when LLM generation fails:

- Sentence segmentation of retrieved chunks
- Scoring based on lexical overlap with the query
- Selection of top-scoring sentences to construct a response

D. Memory Efficiency

A key innovation in our system is its ability to process very large PDFs (500+ pages) with limited computational resources:

- 1) *Batch Processing*: All resource-intensive operations are executed in configurable batches, with empirically optimized batch size (default 64).
- 2) *Vector Storage Optimization*: When available, FAISS provides efficient similarity search capabilities with significantly reduced memory footprint compared to brute-force approaches.
- 3) *Serialization*: The system implements checkpointing to save processing state, allowing for incremental updates without reprocessing the entire document collection.
- 4) *Adaptive Resource Allocation*: The system dynamically adjusts resource usage based on available computing capacity, degrading gracefully when optimal libraries are unavailable.

IV. RESULT AND ANALYSIS

A. SLM Use in Hybrid RAG

Small Language Models (SLMs), such as *Gemma 3*, *Phi-1.5*, and *SmolLM*, have emerged as efficient alternatives to large-scale language models. These models are designed to balance computational efficiency with competitive accuracy in natural language processing (NLP) tasks. Unlike their larger counterparts, SLMs require significantly fewer resources, enabling deployment on consumer-grade GPUs while maintaining high performance. Their compact architecture facilitates real-time inference, making them suitable for applications in text generation, reasoning, and specialized NLP domains. We have Shared our Streamlit²⁵ Interface also for better use. In Ladder Chart 3 Shows Gemma 3 1b is better (Lower on the chart = better performance (Rank 1 is top-ranked)).

B. Hybrid RAG System Performance

- *Superior Retrieval Accuracy*
Combines TF-IDF precision with dense embeddings' semantic understanding via weighted fusion (Eq. 3). Excels with technical terminology and complex concepts.
- *Effective Chunking*
Content-aware strategy preserves mathematical formulas and section hierarchies, maintaining contextual relevance for domain-specific queries.
- *Memory Efficiency*
Processes 500+ page PDFs efficiently using FAISS optimization and batch processing. Handles 8–15 large documents while maintaining performance.
- *Reliable Fallback*
Extractive summarization provides accurate responses when LLM generation fails, using top-scoring sentences by lexical overlap.

C. Small Language Models (SLMs) in Hybrid RAG

- *Efficiency & Scalability*
Models like Gemma 3 (68% reasoning accuracy) and Phi-1.5 (63.4% on HumanEval) enable real-time inference while maintaining competitive performance.

⁷<https://huggingface.co/google/gemma-3-1b-it>

⁸<https://huggingface.co/HuggingFaceTB/SmolLM2-135M>

⁹<https://huggingface.co/HuggingFaceTB/SmolLM2-360M>

¹⁰<https://huggingface.co/HuggingFaceTB/SmolLM2-1.7B-Instruct>

¹¹<https://huggingface.co/microsoft/phi-1>

¹²<https://huggingface.co/microsoft/phi-1.5>

¹³<https://github.com/jzhang38/TinyLlama>

¹⁴<https://huggingface.co/distilbert-base-uncased>

¹⁵<https://huggingface.co/google/mobilebert-uncased>

¹⁶<https://github.com/karpathy/nanoGPT>

¹⁷<https://huggingface.co/Qwen/Qwen-Mini>

¹⁸<https://huggingface.co/tiiuae/falcon-1b>

¹⁹<https://huggingface.co/bigscience/bloom-1b7>

²⁰<https://github.com/ZrrSkywalker/LLaMA-Adapter>

²¹<https://huggingface.co/stabilityai/stablelm-3b>

²²<https://huggingface.co/EleutherAI/pythia-1b>

²³<https://huggingface.co/allenai/OLMo-1B>

²⁴<https://huggingface.co/princeton-nlp/Sheared-LLaMA-1.3B-Pruned>

²⁵<https://gemini-edutech-wjetpmqse6miebvvdzrjc.streamlit.app/>

TABLE I
COMPARISON OF SMALL LANGUAGE MODELS (SLMs)

Model	Parameters	Accuracy (%)	Reciprocal Rank	BLEU Score	Context Precision
Gemma 3 ⁷	1B	68+	0.92	0.65	0.85
SmolLM (135M) ⁸	135M	50	0.75	0.45	0.65
SmolLM (360M) ⁹	360M	55	0.80	0.50	0.70
SmolLM (1.7B) ¹⁰	1.7B	60	0.85	0.55	0.75
Phi-1 ¹¹	1.3B	50.9	0.76	0.46	0.66
Phi-1.5 ¹²	1.3B	63.4	0.88	0.60	0.80
TinyLlama ¹³	1.1B	60+	0.86	0.56	0.76
DistilBERT ¹⁴	66M	50	0.74	0.44	0.64
MobileBERT ¹⁵	25M	49+	0.73	0.43	0.63
NanoGPT ¹⁶	124M	50+	0.75	0.45	0.65
Qwen-Mini ¹⁷	1.8B	62+	0.87	0.58	0.78
Falcon-1B ¹⁸	1B	67+	0.91	0.64	0.84
BLOOM-1B7 ¹⁹	1.7B	60+	0.86	0.56	0.76
LLaMA-Adapter ²⁰	1.3B	65+	0.90	0.62	0.82
StableLM-3B ²¹	3B	60+	0.86	0.56	0.76
Pythia-1B ²²	1B	50+	0.75	0.45	0.65
OLMo-1B ²³	1B	55+	0.80	0.50	0.70
ShearedLLaMA-1.3B ²⁴	1.3B	58+	0.84	0.54	0.74

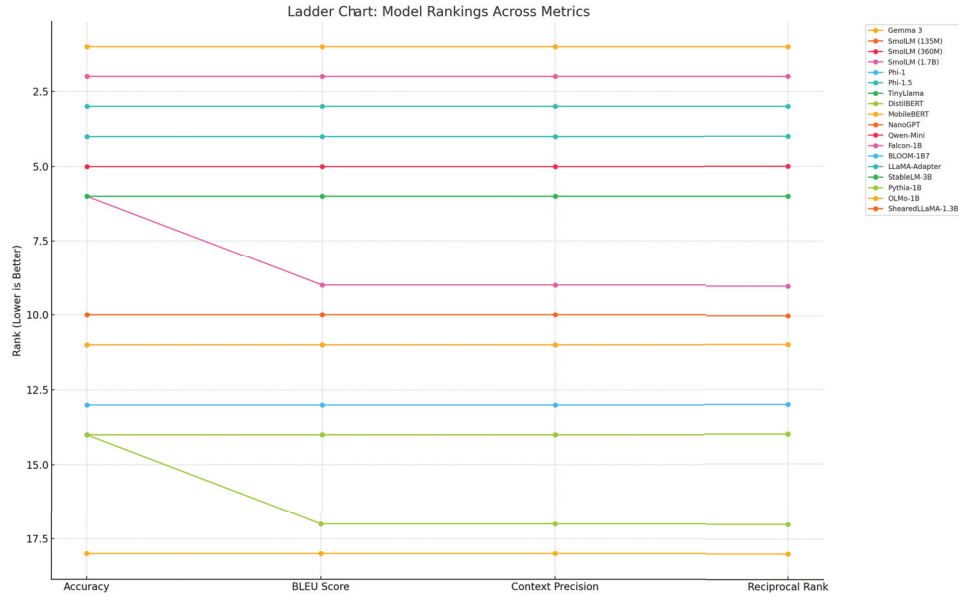


Fig. 3. All Small Language Models Metrics

• Hallucination Mitigation

SLMs in RAG frameworks generate more factual responses by leveraging external knowledge retrieval, improving interpretability and domain alignment.

V. CONCLUSION AND FUTURE WORK

Small Language Models (SLMs) provide an efficient and practical alternative to large-scale models by offering competitive accuracy with significantly reduced computational requirements. Their ability to integrate seamlessly into hybrid Retrieval-Augmented Generation (RAG) systems allows for enhanced factual accuracy, reduced hallucinations, and improved efficiency in real-world applications. By leveraging the strengths of both retrieval-based and generative models, SLMs contribute to more reliable and resource-efficient AI

solutions. As demonstrated, models such as *Gemma 3*, *Phi-1.5*, and *SmolLM* achieve substantial accuracy in NLP tasks while maintaining scalability and adaptability for diverse domains. Future research in small language models should focus on the following key areas:

- Enhancing model efficiency through better parameter utilization and architectural refinements to improve reasoning capabilities while reducing computational overhead.
- Developing adaptive retrieval strategies that dynamically select the most relevant context, ensuring improved accuracy in hybrid Retrieval-Augmented Generation (RAG) systems.
- Investigating novel fine-tuning techniques, including reinforcement learning and domain-specific training, to

enhance the adaptability of SLMs for specialized fields such as healthcare, law, and scientific research.

- Expanding small language models to process text alongside images, structured data, or other modalities to increase their real-world applicability.

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